PADHAKU
Mathematics · Class 10 Mathematics

CHAPTER 9

Some Applications of Trigonometry

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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❓ In-text questions · 9.1 Heights and Distances

? Identify the lines of sight and the angles so formed in Fig. 8.3 (girl and her mother looking at the hot-air balloon). Are they angles of elevation or angles of depression?

✔ ANSWER

In Fig. 8.3 the girl and her mother look up from the ground at the balloon. The line of sight is the line from their eyes to the balloon.

The balloon is above the horizontal level, so they raise their heads to see it.

Both angles are angles of elevation.

? If you want to find the height CD of the minar (Fig. 9.1) without measuring it, what information do you need?

✔ ANSWER

We need three things:

- (i) the distance DE of the student from the foot of the minar,
- (ii) the angle of elevation ∠BAC of the top of the minar,
- (iii) the height AE of the student.

Then CD = BC + BD, where BD = AE (height of the student).

? **In Δ ABC, the side BC is the opposite side to the known ∠A. Which trigonometric ratio can we use to find BC?**

✔ ANSWER

We know AB (= DE, the distance) and ∠A; we need BC.

The ratio that uses AB and BC is tan A (or cot A):

tan A = BC/AB (or cot A = AB/BC), which gives BC = AB × tan A.

Height of the minar = BC + AE.

Exercise 9.1

1 **A circus artist is climbing a 20 m long rope, tightly stretched and tied from the top of a vertical pole to the ground. Find the height of the pole, if the angle made by the rope with the ground level is 30°.**

✔ ANSWER

Let the pole be AB (height h) and the rope AC = 20 m, ∠ACB = 30°.

In right Δ ABC, sin 30° = AB/AC

1/2 = h/20

h = 20/2 = 10

Height of the pole = 10 m

2 **A tree breaks due to storm and the broken part bends so that the top of the tree touches the ground making an angle 30° with it. The distance between the foot of the tree to the point where the top touches the ground is 8 m. Find the height of the tree.**

✔ ANSWER

Let AB be the standing part (height h) and AC the broken part (length l), touching the ground at C, BC = 8 m, ∠ACB = 30°.

tan 30° = AB/BC → 1/√3 = h/8 → h = 8/√3 m

cos 30° = BC/AC → √3/2 = 8/l → l = 16/√3 m

Height of the tree = h + l = 8/√3 + 16/√3 = 24/√3 = 8√3

Height of the tree = 8√3 m (≈ 13.86 m)

3 **A contractor plans to install two slides. For children below 5 years, a slide whose top is at a height of 1.5 m, inclined at 30° to the ground; for elder children, a steep slide at a height of 3 m, inclined at 60° to the ground. What should be the length of the slide in each case?**

✔ ANSWER

Slide for small children: height 1.5 m, angle 30°, length l_1

sin 30° = $1.5/l_1$ → $1/2 = 1.5/l_1$ → l_1 = 3 m

Slide for elder children: height 3 m, angle 60°, length l_2

sin 60° = $3/l_2$ → $\sqrt{3}/2 = 3/l_2$ → $l_2 = 6/\sqrt{3} = 2\sqrt{3}$ m

Lengths: 3 m and $2\sqrt{3}$ m (≈ 3.46 m)

4 **The angle of elevation of the top of a tower from a point on the ground, which is 30 m away from the foot of the tower, is 30°. Find the height of the tower.**

✔ ANSWER

Let the tower be AB (height h), point C on the ground, BC = 30 m, ∠ACB = 30°.

tan 30° = AB/BC

$1/\sqrt{3} = h/30$

$h = 30/\sqrt{3} = 10\sqrt{3}$

Height of the tower = $10\sqrt{3}$ m (≈ 17.32 m)

5 **A kite is flying at a height of 60 m above the ground. The string attached to the kite is temporarily tied to a point on the ground. The inclination of the string with the ground is 60°. Find the length of the string, assuming that there is no slack in the string.**

✔ ANSWER

Let the kite be at A, height AB = 60 m, string AC = l, ∠ACB = 60°.

sin 60° = AB/AC

$\sqrt{3}/2 = 60/l$

$l = 120/\sqrt{3} = 40\sqrt{3}$

Length of the string = 40√3 m (≈ 69.28 m)

6 **A 1.5 m tall boy is standing at some distance from a 30 m tall building. The angle of elevation from his eyes to the top of the building increases from 30° to 60° as he walks towards the building. Find the distance he walked towards the building.**

✔ ANSWER

Height of the building above the boy's eyes = 30 − 1.5 = 28.5 m.

Let the first position be at distance x_1 and the second at x_2 from the building (measured from eye level).

$\tan 30° = 28.5/x_1 \rightarrow 1/\sqrt{3} = 28.5/x_1 \rightarrow x_1 = 28.5\sqrt{3}$ m

$\tan 60° = 28.5/x_2 \rightarrow \sqrt{3} = 28.5/x_2 \rightarrow x_2 = 28.5/\sqrt{3} = 9.5\sqrt{3}$ m

Distance walked = $x_1 - x_2 = 28.5\sqrt{3} - 9.5\sqrt{3} = 19\sqrt{3}$

Distance walked = 19√3 m (≈ 32.91 m)

7 **From a point on the ground, the angles of elevation of the bottom and the top of a transmission tower fixed at the top of a 20 m high building are 45° and 60° respectively. Find the height of the tower.**

✔ ANSWER

Let the building be AB = 20 m, tower BC = h, point P on the ground at distance x from A.

tan 45° = AB/PA → 1 = 20/x → x = 20 m

tan 60° = AC/PA → √3 = (20 + h)/20

20 + h = 20√3

h = 20√3 − 20 = 20(√3 − 1)

Height of the tower = 20(√3 − 1) m (≈ 14.64 m)

8 **A statue, 1.6 m tall, stands on the top of a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and from the same point the angle of elevation of the top of the pedestal is 45°. Find the height of the pedestal.**

✔ ANSWER

Let the pedestal be AB = h, statue BC = 1.6 m, point P at distance x from A.

tan 45° = AB/PA → 1 = h/x → x = h

tan 60° = AC/PA → √3 = (h + 1.6)/h

√3 h = h + 1.6 → h(√3 − 1) = 1.6

h = 1.6/(√3 − 1) = 1.6(√3 + 1)/[(√3 − 1)(√3 + 1)] = 1.6(√3 + 1)/2

Height of the pedestal = 0.8(√3 + 1) m (≈ 2.19 m)

9 **The angle of elevation of the top of a building from the foot of the tower is 30° and the angle of elevation of the top of the tower from the foot of the building is 60°. If the**

tower is 50 m high, find the height of the building.

✔ ANSWER

Let the tower be AB = 50 m, building CD = h, distance between them BD = x.

From the foot of the building D: tan 60° = AB/BD → √3 = 50/x → x = 50/√3 m

From the foot of the tower B: tan 30° = CD/BD → 1/√3 = h/x

h = x/√3 = 50/(√3 × √3) = 50/3

Height of the building = 50/3 m = 16⅔ m (≈ 16.67 m)

10 **Two poles of equal heights are standing opposite each other on either side of the road, which is 80 m wide. From a point between them on the road, the angles of elevation of the top of the poles are 60° and 30°, respectively. Find the height of the poles and the distances of the point from the poles.**

✔ ANSWER

Let the poles be AB and CD, each of height h, and P a point on the road with BP = x, so PD = 80 − x.

In Δ ABP: tan 60° = h/x → h = √3 x ... (1)

In Δ CDP: tan 30° = h/(80 − x) → h = (80 − x)/√3 ... (2)

From (1) and (2): √3 x = (80 − x)/√3 → 3x = 80 − x → 4x = 80 → x = 20

h = √3 × 20 = 20√3

Height of the poles = 20√3 m (≈ 34.64 m)

Distances of the point from the poles = 20 m and 80 − 20 = 60 m

11 **A TV tower stands vertically on a bank of a canal. From a point on the other bank directly opposite the tower, the angle of elevation of the top of the tower is 60°. From another point 20 m away from this point on the line joining this point to the foot of the tower, the angle of elevation of the top of the tower is 30°. Find the height of the tower and the width of the canal.**

✔ ANSWER

Let the tower be AB = h, C the point directly opposite on the other bank, BC = width = w, and D the point 20 m behind C, so BD = w + 20.

In Δ ABC: tan 60° = h/w → h = √3 w ... (1)

In Δ ABD: tan 30° = h/(w + 20) → h = (w + 20)/√3 ... (2)

√3 w = (w + 20)/√3 → 3w = w + 20 → 2w = 20 → w = 10

h = √3 × 10 = 10√3

Height of the tower = 10√3 m (≈ 17.32 m), **width of the canal = 10 m**

12 **From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 45°. Determine the height of the tower.**

✔ **ANSWER**

Let the building be AB = 7 m and the tower CD. Draw AE ⊥ CD, so AE = BC (distance between them) and EC = AB = 7 m.

Angle of depression of C from A = 45° = ∠ACB (alternate angles).

In Δ ABC: tan 45° = AB/BC → 1 = 7/BC → BC = 7 m, so AE = 7 m

In Δ AED: tan 60° = DE/AE → √3 = DE/7 → DE = 7√3 m

Height of the tower CD = DE + EC = 7√3 + 7

Height of the tower = 7(√3 + 1) m (≈ 19.12 m)

13 **As observed from the top of a 75 m high lighthouse from the sea-level, the angles of depression of two ships are 30° and 45°. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships.**

✔ **ANSWER**

Let the lighthouse be AB = 75 m, ships at C (nearer) and D, on the same side.

Angles of depression 45° and 30° equal the angles of elevation from the ships (alternate angles): ∠ACB = 45°, ∠ADB = 30°.

tan 45° = AB/BC → 1 = 75/BC → BC = 75 m

tan 30° = AB/BD → 1/√3 = 75/BD → BD = 75√3 m

Distance between the ships = CD = BD − BC = 75√3 − 75

Distance = 75(√3 − 1) m (≈ 54.9 m)

14 **A 1.2 m tall girl spots a balloon moving with the wind in a horizontal line at a height of 88.2 m from the ground. The angle of elevation of the balloon from the eyes of the girl at any instant is 60°. After some time, the angle of elevation reduces to 30°. Find the distance travelled by the balloon during the interval.**

✔ ANSWER

Height of the balloon above the girl's eyes = 88.2 − 1.2 = 87 m.

Let the horizontal distances from the girl's eyes be x_1 (at 60°) and x_2 (at 30°).

tan 60° = $87/x_1$ → √3 = $87/x_1$ → $x_1 = 87/\sqrt{3} = 29\sqrt{3}$ m

tan 30° = $87/x_2$ → $1/\sqrt{3} = 87/x_2$ → $x_2 = 87\sqrt{3}$ m

Distance travelled = $x_2 - x_1 = 87\sqrt{3} - 29\sqrt{3} = 58\sqrt{3}$

Distance travelled by the balloon = 58√3 m (≈ 100.46 m)

15 **A straight highway leads to the foot of a tower. A man standing at the top of the tower observes a car at an angle of depression of 30°, which is approaching the foot of the tower with a uniform speed. Six seconds later, the angle of depression of the car is found to be 60°. Find the time taken by the car to reach the foot of the tower from this point.**

✔ ANSWER

Let the tower be AB = h. The car is at C when the angle of depression is 30° and at D when it is 60° (D nearer to B).

Angles of elevation from the car: ∠ACB = 30°, ∠ADB = 60° (alternate angles).

tan 30° = h/BC → BC = h√3

tan 60° = h/BD → BD = h/√3

Distance covered in 6 s = CD = BC − BD = h√3 − h/√3 = (3h − h)/√3 = 2h/√3

Speed = (2h/√3) ÷ 6 = h/(3√3) per second

Time to cover the remaining distance BD = (h/√3) ÷ (h/(3√3)) = 3

Time taken = 3 seconds

🎉 **Chapter 9 done! Revise the answers once more tomorrow.**